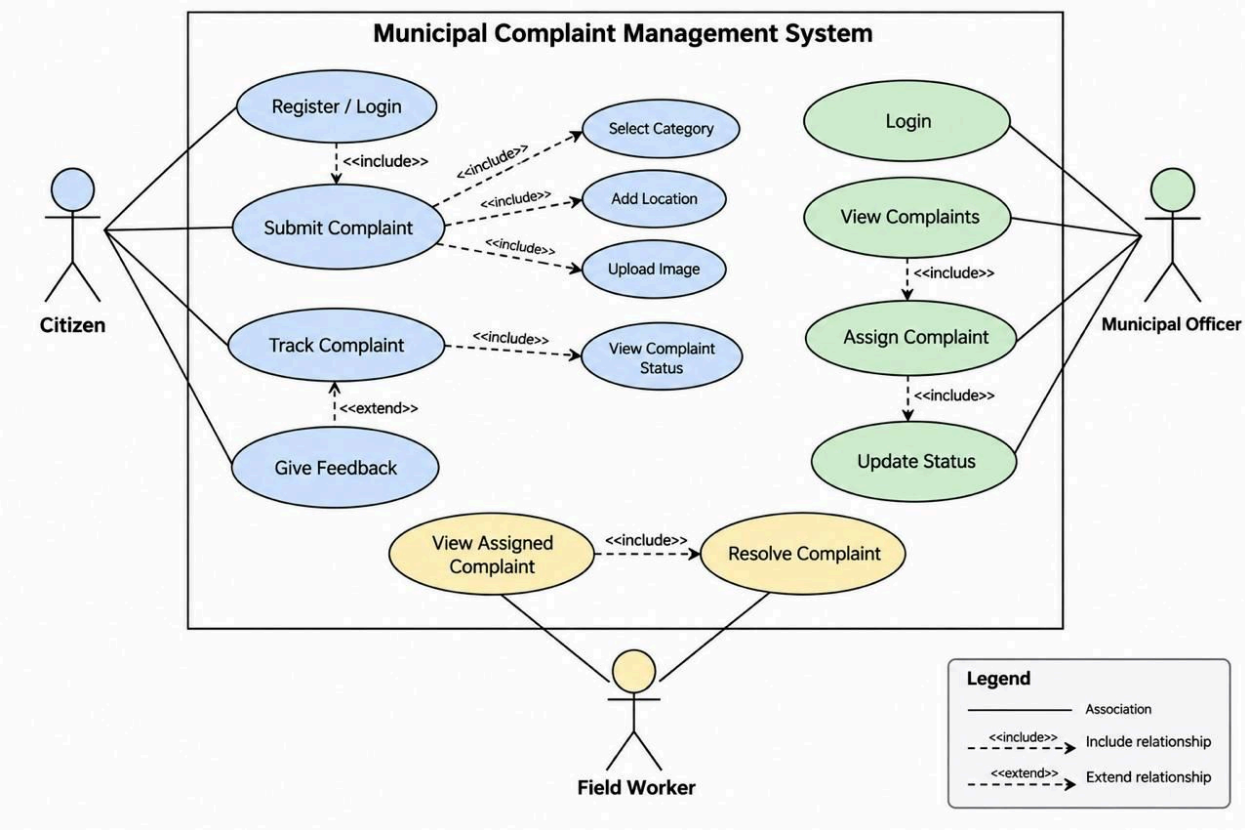
Lab 5 UML Use Case Modeling —



Use Case Specifications

1. Use Case ID: UC-01

- **Use Case Name:** Register / Login
- **Primary Actor:** Citizen
- **Relationship / Type:** Included in UC-02
- **Pre-condition:** Internet access and registered user credentials.
- **Post-condition:** Authenticated user session created.

2. Use Case ID: UC-02

- **Use Case Name:** Submit Complaint
- **Primary Actor:** Citizen
- **Relationship / Type:** Base Use Case (Includes UC-01, UC-03, UC-04, UC-05)
- **Pre-condition:** Citizen is logged into the portal.
- **Post-condition:** Complaint saved and a unique Tracking ID is generated.

3. Use Case ID: UC-03

- **Use Case Name:** Select Category
- **Primary Actor:** Citizen
- **Relationship / Type:** Included in UC-02
- **Pre-condition:** Complaint submission form is open.
- **Post-condition:** Respective department tag is mapped.

4. Use Case ID: UC-04

- **Use Case Name:** Add Location
- **Primary Actor:** Citizen
- **Relationship / Type:** Included in UC-02
- **Pre-condition:** Complaint submission form is open.
- **Post-condition:** Geographic address or GPS coordinates are saved.

5. Use Case ID: UC-05

- **Use Case Name:** Upload Image
- **Primary Actor:** Citizen
- **Relationship / Type:** Included in UC-02
- **Pre-condition:** Complaint submission form is open.
- **Post-condition:** Photo proof is attached to the ticket.

6. Use Case ID: UC-06

- **Use Case Name:** Track Complaint
- **Primary Actor:** Citizen
- **Relationship / Type:** Base Use Case (Includes UC-07)
- **Pre-condition:** Valid Tracking ID exists.
- **Post-condition:** Real-time lifecycle stage is displayed.

7. Use Case ID: UC-07

- **Use Case Name:** View Complaint Status
- **Primary Actor:** Citizen
- **Relationship / Type:** Included in UC-06
- **Pre-condition:** Grievance record exists in the database.
- **Post-condition:** Status details are retrieved and displayed.

8. Use Case ID: UC-08

- **Use Case Name:** Give Feedback
- **Primary Actor:** Citizen
- **Relationship / Type:** Extends UC-06 (`<<extend>>`)
- **Pre-condition:** Complaint status is marked as 'Resolved'.
- **Post-condition:** Rating and feedback comments are recorded.

9. Use Case ID: UC-09

- **Use Case Name:** Login (Officer)
- **Primary Actor:** Municipal Officer
- **Relationship / Type:** Base Use Case
- **Pre-condition:** Valid municipal administrative credentials.
- **Post-condition:** Access granted to officer dashboard.

10. Use Case ID: UC-10

- **Use Case Name:** View Complaints
- **Primary Actor:** Municipal Officer
- **Relationship / Type:** Base Use Case (Includes UC-11, UC-12)
- **Pre-condition:** Officer is authenticated.
- **Post-condition:** Jurisdictional complaint list is rendered.

11. Use Case ID: UC-11

- **Use Case Name:** Assign Complaint
- **Primary Actor:** Municipal Officer
- **Relationship / Type:** Included in UC-10
- **Pre-condition:** Unassigned complaint is selected.
- **Post-condition:** Field worker is linked to the ticket.

12. Use Case ID: UC-12

- **Use Case Name:** Update Status
- **Primary Actor:** Municipal Officer
- **Relationship / Type:** Included in UC-10
- **Pre-condition:** Complaint is reviewed by officer.
- **Post-condition:** Updated status is committed to database.

13. Use Case ID: UC-13

- **Use Case Name:** View Assigned Complaint
- **Primary Actor:** Field Worker
- **Relationship / Type:** Base Use Case (Includes UC-14)
- **Pre-condition:** Field Worker is logged into system.
- **Post-condition:** List of allocated work orders is displayed.

14. Use Case ID: UC-14

- **Use Case Name:** Resolve Complaint
- **Primary Actor:** Field Worker
- **Relationship / Type:** Included in UC-13
- **Pre-condition:** Physical on-site repair is completed.

- **Post-condition:** Complaint marked as completed/resolved.